

De-Americanization

Removing the early-exercise premium from listed option quotes, fast

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Abstract

Listed single-stock and ETF options are *American*: they may be exercised early, so their prices carry an early-exercise premium over the European value. Feeding those prices straight into a European implied-volatility inversion biases the smile — upward, and worst where vega is small. This note documents the Vol-Fitter’s de-Americanization: for each quote it finds the volatility σ^* such that a Cox–Ross–Rubinstein American tree reprices the quote, and returns σ^* as the European-equivalent Black volatility. We explain why this tree inversion is the right control variate, how discrete cash dividends enter through an escrowed-spot lattice, and the numerical choices — tree depth, the bracketed-bisection sweep — that hold the inversion to sub-vol-bp accuracy. The hot path is a Numba kernel that de-Americanizes a whole chain with no per-step allocations: on a 300-quote chain it runs in 0.06 ms versus 12.5 ms for the NumPy batch, a $218\times$ speed-up, and a content-digest cache ensures fit-tuning never re-runs it. Because the inversion is done per strike independently, the European-equivalent call curve it returns can come out slightly *non-convex* in the sparse wings — a butterfly arbitrage in the input — so we also document a cross-strike *convexity repair*, confined to the wings and to the quoted bid–ask band so that the dense at-the-money core is left byte-identical. A worked example shows the naive European inversion over-stating implied volatility by up to 282 volatility basis points on a dividend-paying name, an error the de-Americanization removes entirely.

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1 Why de-Americanize

The volatility models of Notes 01–04 all assume *European* prices: they work with $C_T(k) = \mathbb{E}^T[(e^{X_T} - e^k)^+]$, a terminal expectation. A listed American option is worth more, because its holder may exercise before expiry; the excess is the *early-exercise premium* (EEP). If one inverts an American price with the European Black formula, the extra value is misattributed to volatility, and the implied vol comes out biased.

Heuristic

The bias is not uniform. It is largest exactly where the European model is least sensitive to volatility — deep in-the-money, where vega is small, so a given dollar of early-exercise premium translates into a large spurious vol move. A naive smile built from American quotes therefore has corrupted wings, the very region the fitter most needs to extrapolate. De-Americanization is not a cosmetic preprocessing step; it is the difference between fitting the market’s volatility and fitting its exercise policy.

Figure 1 makes this concrete on American calls written on a stock paying a 6% dividend yield (where early exercise to capture dividends is rational): an option priced at a true 25% volatility, inverted naively as European, implies a volatility up to 282 vol bps too high, rising as the option goes in the money. De-Americanization recovers a flat 25%.

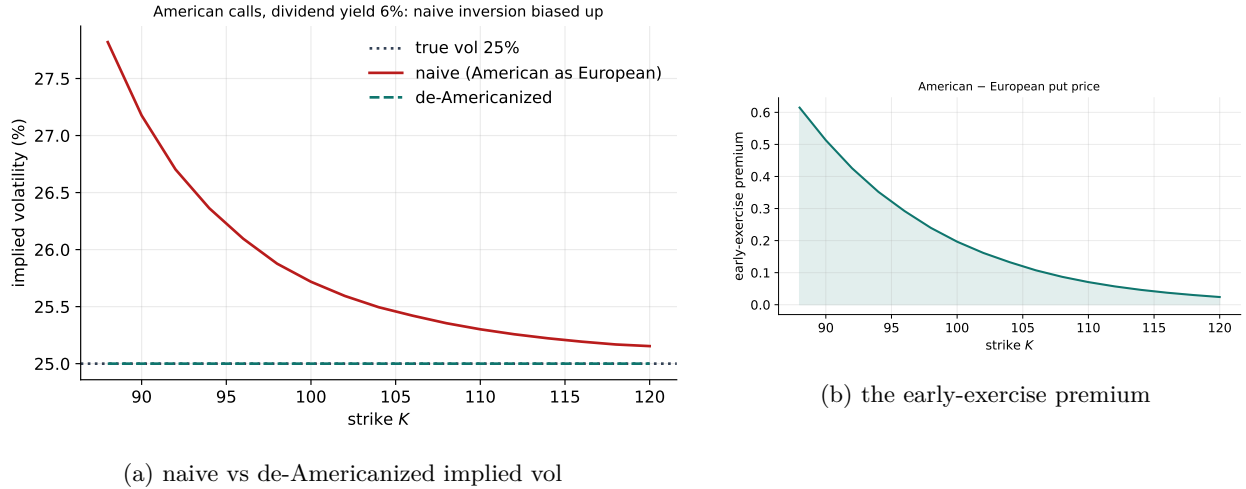


Figure 1: The naive European inversion of an American price is biased up by the early-exercise premium; the de-Americanization returns the true volatility.

2 The pricing tree

2.1 Cox–Ross–Rubinstein

The American value is computed by a Cox–Ross–Rubinstein (CRR) binomial tree. With N steps of size $\Delta t = T/N$, the spot moves by $u = e^{\sigma\sqrt{\Delta t}}$ or $d = 1/u$ each step, with risk-neutral up-probability

$$p = \frac{e^{(r-q)\Delta t} - d}{u - d}, \quad 0 < p < 1, \quad (1)$$

(r the rate, q a continuous dividend yield). The condition $0 < p < 1$ is the CRR no-arbitrage requirement $d < e^{(r-q)\Delta t} < u$; it fails only for extreme drift-to-vol ratios and then *raises* rather than silently mis-pricing. The tree rolls back from the payoff with discounting $e^{-r\Delta t}$, taking $\max(\text{continuation}, \text{intrinsic})$ at each node for the American feature. The European value is the same recursion without the intrinsic comparison. Stripped of the escrowed-dividend machinery, the kernel is a dozen lines (this reproduces `binomial_price` at $q = 0$ to 10^{-10} ; the European leg matches Black–Scholes to 2×10^{-3} at depth):

Listing 1: The CRR backward induction with early exercise (distilled from `core/american.py`).

```

1 def crr_price(is_call, s, k, t, sigma, r, n=501, american=True):
2     dt = t/n; u = np.exp(sigma*np.sqrt(dt)); d = 1/u
3     p = (np.exp(r*dt) - d)/(u - d); disc = np.exp(-r*dt) # risk-
        neutral up-prob
4     sT = s * u**np.arange(-n, n+1, 2) # terminal
        spots
5     v = np.maximum(sT-k, 0.0) if is_call else np.maximum(k-sT, 0.0)
6     for m in range(n-1, -1, -1): # roll back
        one layer
7         v = disc*(p*v[1:] + (1-p)*v[:-1]) #
        continuation value
8         if american: # early-
            exercise test:
9             sM = s * u**np.arange(-m, m+1, 2) # max(
                continuation,
10            v = np.maximum(v, sM-k if is_call else k-sM) #
                intrinsic)
11     return v[0]
```

2.2 Discrete cash dividends

A continuous yield q is a crude proxy for a real single name. Discrete cash dividends are handled by the *escrowed-spot* method: the present value of dividends with ex-dates in $(0, T]$ is stripped from the spot to form the diffusing “base” lattice, and the actual spot at each node is the base lattice plus the present value of dividends not yet paid. This keeps the lattice recombining (cash dividends otherwise break recombination) while pricing early exercise against the *actual* cum-dividend spot. The two mechanisms coexist: discrete cash for near dividends, a continuous proxy for the far tail.

3 The de-Americanization map

3.1 Tree inversion as a control variate

De-Americanization is a one-dimensional root find. Given an American quote A at (s, K, T, r, q) ,

$$\boxed{\text{find } \sigma^* : \text{CRR}_{\text{Am}}(\sigma^*) = A, \quad \text{return } \sigma^* \text{ as the European-equivalent Black vol.}} \quad (2)$$

Heuristic

Why is the *tree's* σ^* the right European volatility? Because the tree is used as a *control variate* for the early-exercise feature. The quantity we ultimately want is the European Black vol; the market price differs from the European price by an EEP we cannot observe. The tree supplies a model EEP as a function of σ . Solving $\text{CRR}_{\text{Am}}(\sigma^*) = A$ implicitly subtracts the model EEP at σ^* , and the European Black price at the *same* σ^* equals the tree's European value (the two agree to tree discretization, $\sim 10^{-5}$ at the default depth). So σ^* is exactly the de-Americanized Black vol: the early-exercise premium cancels in the control variate, leaving the diffusion volatility. The method is exact in the model and robust in practice because the EEP is a smooth, slowly varying function of σ .

3.2 Static bounds and bracketing

A price below intrinsic (impossible for an American option) or above the static upper bound (s for a call, K for a put) yields `nan`, mirroring the fitter's convention for unusable quotes. The bracket lower end is lifted just above the CRR drift floor $\sigma\sqrt{\Delta t} > |r - q|\Delta t$ so (1) stays valid, and the upper end is expanded until the price is bracketed, capped at $\sigma = \text{SIGMA_HI} = 4$.

4 Numerical choices

4.1 Depth and bisection

Two depths are used. The *scalar* path (one quote at a time) uses $N = \text{DEFAULT_STEPS} = 501$ and a Brent root find; the European leg of the tree then matches Black to $\sim 10^{-5}$. The *batch* path uses a coarser $N = 192$ tree and a fixed 24-sweep bracketed bisection, which prices within a few 10^{-4} of Black and resolves σ^* to ~ 0.002 vol bps — far below the fit budget. The bisection count was tuned down from 45 to 24 at no measurable accuracy cost (< 0.01 vol bp drift), a $\sim 1.8\times$ saving on the isolated de-Am rail.

Caution

The tree depth is a numerical-target knob, not a free speed dial. Cutting the batch depth from 192 to 128 is faster (the backward induction is $O(N^2)$, so $\sim 2\times$) but shifts the priced European leg by ~ 15 vol bps — i.e. it changes the *answer*, not just the runtime. The depth is held at 192 and speed is bought elsewhere (the Numba kernel, section 4.2).

4.2 The Numba kernel

The hot path de-Americanizes a whole chain. The NumPy batch vectorizes across quotes but still allocates per step; the compiled kernel does a single scalar CRR per quote with no per-step allocation — precomputed power tables, the escrow lattice constants hoisted out, and a `prange` parallel loop over quotes.

Performance

On a 300-quote chain the Numba kernel runs in 0.06 ms versus 12.5 ms for the NumPy batch — a $218\times$ speed-up (the documented figure on wide real chains is $\sim 60\times$; this synthetic chain, being uniform, parallelizes especially well). If Numba is unavailable the code falls back to the NumPy batch with identical results to bisection rounding. The kernel agrees with the scalar tree to tree-rounding precision.

4.3 The content-digest cache

De-Americanization depends only on the de-Am *inputs* — forward, discount, the cash-dividend schedule, the year fractions, and the as-of — not on any fit setting. The pipeline keys a cache on a content digest of exactly those inputs, so tuning a fit penalty, changing the model, or re-weighting quotes never re-runs the trees. The de-Am cost is paid once per genuine data change and is then free across the entire calibration sweep. A conservative pre-screen (non-positive bids, a wing buffer) drops quotes that cannot survive the static bounds *before* the tree runs, and is output-preserving by construction.

5 Cross-strike convexity: repairing the de-Americanized wings

De-Americanization inverts each strike *independently* — its own root find (2), its own $\max(\text{EEP}, 0)$ clamp — with no coupling across strikes. That independence is what makes the Numba kernel embarrassingly parallel (section 4.2), but it carries a subtle cost: the set of European-equivalent call prices $\{C(K_i)\}$ it returns need not be *convex* in the strike.

Convexity is not optional. Writing the undiscounted call as $C(K) = \mathbb{E}^{\mathbb{Q}}[(S_T - K)^+]$, its second derivative in strike is the risk-neutral density, $\partial^2 C / \partial K^2 = f_{S_T}(K) \geq 0$: *convexity of the call curve in strike is exactly the no-butterfly-arbitrage condition*. On a discrete strike grid the test is the divided second difference

$$\Delta_i^2 = \frac{C_{i-1}(K_{i+1} - K_i) - C_i(K_{i+1} - K_{i-1}) + C_{i+1}(K_i - K_{i-1})}{K_{i+1} - K_{i-1}} \geq 0, \quad (3)$$

non-negative iff C is convex at K_i . Because de-Am strips a one-sided, strike-local premium in the sparse, low-vega wings, Δ_i^2 can come out slightly negative there — a genuine (small) butterfly arbitrage in the *input* handed to every downstream model, and the seed of the Multi-Core SIV put-wing violation of Note 03.

Caution

The reverted global fix. The first repair projected the *whole* call curve onto the convex cone with a free affine part. Repairing a wing dip that way re-tilted the affine baseline and moved the *at-the-money* call price by a fraction of a cent — negligible in price, but enormous in ATM implied volatility, where vega is largest. It surfaced live as an ATM smile gap on SPY and NVDA and was reverted. The lesson: the dense, high-vega ATM core must be held *exactly* fixed; only the wings may move.

The shipped repair is therefore *wing-only and band-constrained*. With standardized moneyness

$z = k/\sqrt{w_{\text{atm}}}$, strikes in the core $|z| \leq Z_{\text{core}} = 1$ are held byte-identical. Each wing, anchored at its core boundary, is replaced by the ℓ^2 -closest curve that is simultaneously *convex* and *inside the quoted band*, i.e. the projection onto the intersection $\mathcal{C} \cap \mathcal{B}$ of

$$\mathcal{C} = \{\hat{C} : \hat{C} \text{ convex, leaving the anchor no flatter than the core}\}, \quad \mathcal{B} = \{\hat{C} : C^{\text{bid}} \leq \hat{C} \leq C^{\text{ask}}\}, \quad (4)$$

computed by Dykstra’s alternating projection — the convex-cone step is a non-negative-hinge least squares, the band step a clip. A common per-strike shift is applied to bid, mid and ask so the *spread is preserved*, and the repaired mids are re-inverted to total variance.

Heuristic

Why the band constraint is load-bearing. Plain convex projection of an illiquid, genuinely non-convex wing is free to push a price all the way to the static no-arb boundary — and a wing call price at that boundary inverts to an absurd implied volatility (a put wing was seen to jump from 27% to 104%, itself Lee-violating), detonating the downstream fit. Confining the repaired price to the *quoted spread* bounds the correction to the market’s own stated uncertainty: the repair can move a price only as far as the bid–ask allows, which is exactly as far as the quote is genuinely ambiguous.

The repair is gated American-only and short-circuits on any curve already convex to a tolerance $\text{CONVEX_TOL} = 10^{-3}$ (normalized price units), calibrated so that dense liquid chains — whose de-Am wings carry only $\sim 10^{-4}$ rounding curvature — are *never* touched (byte-identical, ATM implied-vol difference $\sim 10^{-15}$), while genuinely arbitrated illiquid wings (10^{-3} to 10^{-2}) are repaired. On those illiquid nodes an ablation found the repair both *reduces* the fitted arbitrage and *lowers* the in-sample error — the model no longer chases the arbitrated quotes — and it composes with the SIV put-wing penalty of Note 03 rather than duplicating it: the de-Am repair removes the arbitrage at the *source*, the penalty enforces the constraint on the *output*, and running both attains the penalty’s arbitrage removal at a fraction of its standalone in-sample cost.

6 Worked example and measured cost

Figure 1 is generated by pricing American calls on a 6%-dividend stock at a true 25% volatility and inverting each both ways. The naive European inversion over-states the volatility by up to 282 vol bps, rising into the money exactly as section 1 predicts; the de-Americanization returns a flat 25%. The early-exercise premium (panel b) is the economic source of the bias: positive, and largest in the money where early exercise to capture the dividend is most valuable. The timing table (table 1) records the kernel speed-up on a wide chain.

Table 1: De-Americanizing a 300-quote chain (fastest of three repeats, 192-step trees, 24 bisections).

Path	time (ms)	relative
NumPy batch	12.5	1×
Numba kernel	0.06	218×

7 What is original, and future directions

The control-variate view of tree inversion is folklore; the contributions here are the *allocation-free Numba kernel* (the per-step allocation, not the arithmetic, was the NumPy bottleneck), the *content-digest cache* that scopes de-Am cost to genuine data changes rather than to fit iterations, the *measured* depth/bisection tuning that cut work without moving the numerical target, and the *wing-only, band-constrained convexity repair* (section 5) that removes the butterfly arbitrage the per-strike inversion leaves in the sparse wings while holding the dense ATM core exactly fixed. The natural next step — documented but not yet shipped — is to seed the bracket with a fast analytic American approximation (Barone–Adesi–Whaley or Bjerk Sund–Stensland 2002) so each tree starts near its root, composing with the kernel for a further constant-factor win; these are *seeds*, not the answer, so they change no contract.

Caution

De-Americanization inherits the model’s assumptions: a flat dividend treatment, no discrete jumps, and the CRR diffusion. For names with hard-to-model dividend timing or borrow, σ^* is only as good as the forward and dividend inputs (Note 06). The method removes the *early-exercise* bias; it cannot repair a wrong forward.

A Hyperparameter atlas

Table 2: De-Americanization hyperparameters (all hidden / internal).

Constant	Default	Role
DEFAULT_STEPS	501	Scalar CRR tree depth; European leg matches Black to $\sim 10^{-5}$.
DEFAULT_BATCH_STEPS	192	Batch/kernel tree depth.
BATCH_BISECTIONS	24	Bracketed-bisection sweeps (~ 0.002 vol bp).
SIGMA_LO	10^{-4}	Lower vol bracket (lifted above the CRR drift floor).
SIGMA_HI	4.0	Upper vol bracket cap.
escrow PV	—	Discrete-cash-dividend escrowed-spot lattice.
pre-screen buffer	$1.5\times$	Wing buffer for the conservative bid/bounds pre-screen.
content digest	—	Cache key (forward, discount, dividends, t , τ , as-of).
convex_deam	true	Enable the wing-only convex de-Am repair (section 5); American chains only, European a no-op.
Z_{core}	1.0	ATM core half-width (units of $\sqrt{w_{\text{atm}}}$) held byte-identical by the repair.
CONVEX_TOL	10^{-3}	Convexity short-circuit: a curve convex to this (normalized price) is left untouched.

B Performance notes

1. **Numba kernel** (section 4.2). Single scalar CRR per quote, no per-step allocation, precomputed powers, `prange` over quotes; measured $218\times$ over the NumPy batch on a 300-quote chain. Graceful NumPy fallback.
2. **Bisection tuning**. $45 \rightarrow 24$ sweeps, < 0.01 vol bp drift, $\sim 1.8\times$ on the isolated rail.
3. **Content-digest cache**. De-Am re-runs only on a genuine data change, never on fit-setting changes — so it is free across a calibration sweep.
4. **Output-preserving pre-screen**. Drops unusable quotes before the tree, by construction not changing any surviving result.
5. **Future (not shipped)**: BAW / BS-2002 analytic bracket seeds to start each tree near its root.

C Reference implementation: the wing convexity repair

The convexity test (3) and the wing repair of section 5 are compact. `min_butterfly` is the divided second difference (its minimum is ≥ 0 iff the curve is convex, i.e. butterfly-free); `convex_fit` is the convex-cone half of the Dykstra projection — the ℓ^2 -closest convex curve leaving the core anchor, written as an affine part plus non-negative hinge knots, so $\delta_j \geq 0$ is exactly the convexity constraint. (The shipped repair alternates this with a bid/ask clip; see section 5.) Both reproduce their production counterparts to 10^{-9} .

Listing 2: The no-butterfly test and the convex wing fit (distilled from `calib/convex_deam.py`).

```

1  def min_butterfly(K, C):                                     #  $\geq 0 \iff C$  convex in
    strike (no bfly)
2      a, b = K[1:-1]-K[:-2], K[2:]-K[1:-1]                 # left / right strike
    gaps
3      fly = C[:-2]*b - C[1:-1]*(a+b) + C[2:]*a             # divided second
    difference
4      return (fly / (a+b)).min()
5
6  def convex_fit(u, c, slope_floor):                          # L2-closest convex curve
    , anchor c[0]
7      cols = [u] + [np.maximum(u-u[j], 0.0) for j in range(1, len(u)-1)]
8      A = np.column_stack(cols)                             #  $[u, (u-u_j)_+ \dots]$  :
    affine + hinges
9      lb = np.zeros(A.shape[1]); lb[0] = slope_floor        # leaving slope
     $\geq$  core;  $\delta_j \geq 0$ 
10     x = lsq_linear(A, c - c[0], bounds=(lb, np.inf)).x
11     return c[0] + A @ x                                     # convex by construction
    ( $\delta_j \geq 0$ )

```

References

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